Exam Center: …….

Room N°: …… **Seat N°:** ……

Full Name: …….

Signature: ……...

Score: 75 marks

Senior High School National Exam

Examination Date: 08 October 2024

Subject: Physics (Science)

Duration: 90 minutes

Topics:

I. (10 pts) a. What is a thermodynamic system?

b. What is the alternating current? What is the difference between alternating current and direct current?

II. (10 pts) A container holds 2.00 moles of an ideal gas at a temperature of 300 K. Calculate:

a. The average kinetic energy of a single gas molecule.

b. The total kinetic energy of all gas molecules.

Given universal gas constant $R = 8.31$ J/mol.K and Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K.

III. (10 pts) The displacement of a wave is described by the equation $y = 1.00 \sin(0.50x - 0.50t)$ m. Calculate the amplitude, wave number, period and speed of the wave.

IV. (15 pts) A diesel engine has a thermal efficiency of 0.50 and receives 400 MJ of heat energy. Calculate:

a. The mechanical work produced by the piston.

b. The amount of heat rejected to the atmosphere.

c. The useful work delivered to the crankshaft if the mechanical efficiency is 90%.

V. (10 pts) A solenoid is 1.00 m long and has 1000 turns. It stores a magnetic energy of 0.4π J when a current of 4 A flows through it. Given that the permeability of free space is $\mu_0 = 4\pi \times 10^{-7}$ Tm/A,

a. Calculate the inductance of the solenoid.

b. Calculate the cross-sectional area of the solenoid.

VI.(15 pts) Between two terminals A and B, a voltage $v = V\sqrt{2}\cos(2\pi f)$ is applied, where $f = 50$ Hz and $V = 65$ V. An inductor with resistance $R_0 = 12\ \Omega$ and unknown inductance L_0 is connected across terminals A and B. The RMS current through the inductor is $I_0 = 5$ A.

a. Calculate the inductance L_0 of the inductor.

b. Write the expression for the instantaneous current $i_0(t)$ as a function of time t, given that $\tan \phi = \phi$.

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Score: 75 marks

Senior High School National Exam
Examination Date: 06 November 2023
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (10 pts) a. State the kinetic theory of ideal gases.
b. What is the study of the laws of thermodynamics about?

II. (10 pts) An ideal gas occupies a volume of V=100 cm^3 at a pressure of $P = 2.00\times10^5\ \mathrm{Pa}$ and a temperature of 27 °C. Calculate the number of moles of the gas.
Given universal gas constant $R = 8.31\ \mathrm{J/mol.K}$.

III. (10 pts) A gas expands from an initial volume of $0.50\ \mathrm{m}^3$ to a final volume of $0.70\ \mathrm{m}^3$, while the pressure increases from $200 \times 10^3\ \mathrm{Pa}$ to $300 \times 10^3\ \mathrm{Pa}$. Calculate the work done by the gas during this expansion.

IV. (15 pts) Every second, a diesel engine receives $200\ \mathrm{kJ}$ of heat energy from the combustion of the fuel mixture and rejects $150\ \mathrm{kJ}$ of heat to the atmosphere. Given that the mechanical efficiency of the engine is 90%, calculate:

a. The mechanical work output over a duration of $1\ \mathrm{minute}$.
b. The overall efficiency of the crankshaft.
c. The useful mechanical work delivered to the crankshaft in $1\ \mathrm{minute}$.
d. The useful efficiency of the engine.

V. (15 pts) A diesel engine has a thermal efficiency of 0.50 and receives $400\ \mathrm{MJ}$ of heat energy. Calculate:
a. The mechanical work produced by the piston.
b. The amount of heat rejected to the atmosphere.
c. The useful work delivered to the crankshaft if the mechanical efficiency is 90%.

VI. (10 pts) An air-core solenoid is $60\ \mathrm{cm}$ long with a diameter of $3.0\ \mathrm{cm}$ and consists of $5000\ \mathrm{turns}$ of copper wire. The copper wire has a diameter of 0.3 mm and a resistivity of $\rho = 1.7\times10^{-8}\ \Omega\cdot\mathrm{m}$. A current of $2.0\ \mathrm{A}$ flows through the solenoid. Given that the permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \mathrm{Tm/A}$, calculate:
a. The magnetic field along the central axis of the solenoid.
b. The total length of copper wire used to wind the solenoid.
c. The electrical resistance of the copper wire.

VII.(15 pts) A capacitor with a capacitance of $C = 10\ \mu\mathrm{F}$ is connected across an alternating voltage source with an RMS voltage of $V_{\mathrm{rms}} = 220\ \mathrm{V}$ and a frequency of $50\ \mathrm{Hz}$.
a. Calculate the impedance of the circuit and the RMS current flowing through the capacitor.
b. Taking the current $i(t) = I\sqrt{2}\sin\omega t$ as the reference, find the expression for the instantaneous voltage $v\ (t)$ across the capacitor.

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Room N°: …… **Seat N°:** ……

Full Name: …….

Signature: ……...

Score: 75 marks

Senior High School National Exam

Examination Date: 05 December 2022

Subject: Physics (Science)

Duration: 90 minutes

Topics:

I. (10 pts) a. What is a thermodynamic process?

b. When does the phenomenon of self-induction occur? Provide an example to illustrate self-induction.

II. (10 pts) The displacement of a wave is described by the equation $y = 0.50 \sin (0.40x - 0.40t)$ m. Calculate the amplitude, wave number, period and speed of the wave.

III. (10 pts) Calculate the volume of a container occupied by 16 g of oxygen gas at a pressure of 1 atm and a temperature of 27 °C. Given universal gas constant $R = 8.31$ J/mol. K and molar mass of the oxygen is 32 g/mole.

IV. (15 pts) An ideal gas consisting of 2.0 moles undergoes a change in volume at a constant temperature of 27 °C. Given: $\ln\left(\frac{10.0}{3.68} = 1\right)$ and universal gas constant $R = 8.31$ J/mol.K. Calculate work done during this volume expansion.

V. (15 pts) An engine has a thermal efficiency of 50% and absorbs 5000 J of heat energy.

Calculate:

a. The mechanical work output from the piston.

b. The amount of heat energy rejected to the surroundings (atmosphere).

c. The useful work delivered to the crankshaft if the mechanical efficiency is 80%.

VI. (15 pts) In an ideal LC circuit undergoing undamped oscillations, the voltage across the capacitor is V_1=200 V when the current through the inductor is zero. Calculate the current in the circuit when the capacitor voltage is V_2 =100 V. Given: Capacitance, $C = 1$ μF and inductance, $L = 0.5$ H.

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Score: 75 marks

Senior High School National Exam
Examination Date: 27 December 2021
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (10 pts) a. What is a wave?

b. What are sources of a magnetic field?

II. (10 pts) Calculate the volume occupied by $28\ \text{g}$ of nitrogen gas in a container at a pressure of $1\ \text{atm}$ and a temperature of $127\ ^\circ\text{C}$. Given: universal gas constant $R = 8.31\ \text{J/mol. K}$ and molar mass of the nitrogen is $28\ \text{g/mole}$.

III. (10 pts) A solenoid is $3.14\ \text{m}$ long and has $500\ \text{turns}$. It stores a magnetic energy of $0.25\ \text{J}$ when an electric current of $10\ \text{A}$ passes through it.

Given: the permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \text{Tm/A}$.

a. Calculate the inductance of the solenoid.

b. Calculate the cross-sectional area of the solenoid.

IV. (15 pts) Calculate the change in internal energy of a system in each case:

a. The system absorbs $1000\ \text{calories}$ of heat energy and performs $800\ \text{J}$ of work.

b. The system absorbs $600\ \text{calories}$ of heat energy and receives $840\ \text{J}$ of work from the surroundings.

c. The system rejects $2400\ \text{calories}$ of heat energy while its volume remains constant.

Given: $1\ \text{calories} = 4.2\ \text{J}$

V. (15 pts) An engine of a modern vehicle has a thermal efficiency of 0.50 and absorbs $5.0\ \text{MJ}$ of heat energy. Calculate:

a. The mechanical work delivered by the piston.

b. The amount of heat energy rejected to the surroundings.

c. The useful work if the mechanical efficiency is 0.90.

VI. (15 pts) A straight conductor of length $l = 30\ \text{cm}$ carries an electric current of $I = 5.0\ \text{A}$ and is placed in a uniform magnetic field of magnetic flux density $B = 3.0\ \text{T}$. Calculate the magnetic force acting on the conductor when it makes the following angles with the magnetic field:

$\theta_1 = 0^\circ,\ \theta_2 = 30^\circ,\ \theta_3 = 60^\circ,\ \theta_4 = 90^\circ, \theta_5 = 180^\circ$.

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Score: 75 marks

Senior High School National Exam

Examination Date: 19 August 2019

Subject: Physics (Science)

Duration: 90 minutes

Topics:

I. (10 pts) a. What is a thermodynamic process?

b. State Lenz's law to determines the direction of the induced current.

II. (10 pts) Calculate the average kinetic energy of a gas molecule at a temperature of $931\ ^\circ\mathrm{C}$. Given: universal gas constant $R = 8.31\ \mathrm{J/mol.\ K}$ and Avogadro number $N_\mathrm{A} = 6.022 \times 10^{23}\ \mathrm{molecules/mole}$.

III. (10 pts) A Carnot engine operates between a cold reservoir and a hot reservoir. If the desired thermal efficiency is 80%, what must be the temperature of the hot reservoir in degrees Celsius?

IV. (15 pts) a. A capacitor with capacitance $C = 1\ \mu\mathrm{F}$ is charged by a voltage $E = V = 1\mathrm{V}$. Calculate the energy stored in the capacitor when it is fully charged.

b. The charged capacitor is then connected to an inductor with inductance $L = 0.1\ \mathrm{H}$ and negligible resistance. Calculate the maximum current i_m in the circuit.

V. (15 pts) Two parallel rails are placed in a horizontal plane, with their ends connected by a $4\ \Omega$ resistor. The rails are separated by a distance of $10\ \mathrm{cm}$. A metal bar MN is placed perpendicular to the rails. The plane of the rails is perpendicular to a uniform magnetic field with magnetic induction $B = 0.05\ \mathrm{T}$. The metal bar MN is forced to move along the rails at a speed of $40\ \mathrm{m/s}$. Calculate the induce current in the metal bar MN.

VI. (15 pts) A thermodynamic system undergoes the following processes. Calculate the change in internal energy ΔU of the system for each case:

a. The system absorbs $2000\ \mathrm{J}$ of heat energy and receives $500\ \mathrm{J}$ of work.

b. The system absorbs $1500\ \mathrm{J}$ of heat and does $400\ \mathrm{J}$ of work.

c. The system rejects $50000\ \mathrm{J}$ of energy while its volume remains constant.

Exam Center: …….
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Full Name: …….
Signature: ……...
Score: 75 marks

Senior High School National Exam
Examination Date: 20 August 2018
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (10 pts) State the kinetic theory of an ideal gas and the first law of thermodynamics.

II.(10 pts) Calculate the volume occupied by $3.2\ \text{g}$ of oxygen gas in a container at a pressure of $1.0 \times 10^5\ \text{Pa}$ and a temperature of $27°\text{C}$. Given: universal gas constant $R = 8.31\ \text{J/mol.K}$.

III. (10 pts) Work of $20\ \text{kJ}$ is done on an enclosed system. Subsequently, $1\ \text{kcal}$ of heat energy is expelled from the system. Calculate the change in the internal energy of the system. Given: $1\ \text{calory} = 4186\ \text{J}$.

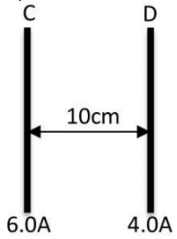
IV. (15 pts) An engine has a thermal efficiency of 0.40 and absorbs $5.0\ \text{MJ}$ of energy. Calculate:

a. The mechanical work produced by the piston.
b. The amount of energy expelled to the atmosphere.
c. The useful work output if the mechanical efficiency is 0.80.

V. (15 pts) Two parallel conductors are separated by a distance of $10\ \text{cm}$ and carry currents of $6.0\ \text{A}$ and $4.0\ \text{A}$, respectively.

Given: The permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \text{T·m}$.

Calculate the magnetic force experienced by the wire D of length $1\ \text{m}$ in the following cases:



a. The currents in both conductor's flow in the same direction.
b. The currents in both conductor's flow in opposite directions.

VI. (15 pts) A solenoid of length $1.5\ \text{m}$ has $470\ \text{turns per meter}$. When a current of $I = 12\ \text{A}$ flows through the solenoid, the magnetic energy stored is $0.31\ \text{J}$. Given: The permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \text{Tm/A}$.

a. Calculate the inductance of the solenoid.
b. Calculate the cross-sectional area of the solenoid.

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Full Name: …….
Signature: ……...
Score: 75 marks

Senior High School National Exam
Examination Date: 21 August 2017
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (5 pts) Calculate the average kinetic energy of gas molecules at a temperature of $1727°\mathrm{C}$. Given: universal gas constant $R = 8.31\ \mathrm{J/mol.\ K}$ and Avogadro number $N_A = 6.022 \times 10^{23}\ \mathrm{molecules/mole}$.

II. (5 pts) The displacement of a wave is described by the equation
$y = 0.50 \sin (0.20x - 0.20\mathrm{t})\ \mathrm{m}$.
Calculate the amplitude, wave number, wavelength, period and speed of the wave.

III. (10 pts) Calculate the change in internal energy ΔU of the system in each of the following cases:

a. The system does $500\ \mathrm{J}$ of work while it expands adiabatically.

b. During adiabatic compression, $1000\ \mathrm{J}$ of work is done on the gas.

IV. (15 pts) A diesel engine in a vehicle has a thermal efficiency of 0.40 and absorbs $6.0 \times 10^6\ \mathrm{J}$ of heat energy. Calculate:
a. The mechanical work produced by the piston.
b. The amount of energy expelled to the atmosphere.
c. The useful work output if the mechanical efficiency is 0.80.

V. (10 pts) A particle with an electric charge of $1.6 \times 10^{-19}\ \mathrm{C}$ moves at a speed of $5.0 \times 10^5\ \mathrm{m/s}$ in a magnetic field of strength $B = 0.26\ \mathrm{T}$. The particle's path curves with a radius of curvature $2.0 \times 10^{-2}\ \mathrm{m}$, and its velocity is perpendicular to the magnetic field.
Calculate the mass of the particle.

VI. (15 pts) Two cylindrical conductor bars are placed on a horizontal plane, with their ends connected by a resistor of resistance $R = 6.0\ \Omega$. The bars are separated by a distance of 30 cm. A metal bar *MN* is placed perpendicular to the two conductor bars. The plane of the conductor bars is perpendicular to a uniform magnetic field with magnetic induction $B = 0.20\ \mathrm{T}$. The metal bar *MN* is forced to move along the two conductor bars at a speed of $100\ \mathrm{m/s}$. Assuming the metal bar and the conductor bars have negligible resistance, calculate the induced current passing through the resistor.

VII. (15 pts) A solenoid has a length of $1.0\ \mathrm{m}$ and $500\ \mathrm{turns}$. When a current of $5\ \mathrm{A}$ passes through it, the magnetic energy stored is $0.5\pi\ \mathrm{J}$.

Given: The permeability of free space is $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{Tm/A}$.

a. Calculate the inductance of the solenoid.
b. Calculate the cross-sectional area of the solenoid.

Exam Center: …….

Room N°: …… **Seat N°:** ……

Full Name: …….

Signature: ……...

Score: 75 marks

Senior High School National Exam

Examination Date: 22 August 2016

Subject: Physics (Science)

Duration: 90 minutes

Topics:

I. (5 pts) What is a thermodynamic process?

II. (5 pts) When an electric current passes through an inductor, it is observed that magnetic field lines emerge from one end and enter the other end of the inductor. Which end corresponds to the north pole and which to the south pole of the inductor?

III. (10 pts) Calculate the volume occupied by $6.4\ \mathrm{g}$ of oxygen gas in a container at a pressure of $1.0\times10^{5}\ \mathrm{Pa}$ and a temperature of $400\ \mathrm{K}$. Given: the molar mass of oxygen, $M = 32\ \mathrm{g/mol}$.

IV. (15 pts) A capacitor with capacitance $C = 2.0\ \mu\mathrm{F}$ is charged to a voltage $V = 5.0\mathrm{V}$. Calculate the electric energy stored in the capacitor.

V. (10 pts) Calculate the change in internal energy ΔU of the system for each case:

a. The system absorbs $2000\ \mathrm{J}$ of heat energy and does $500\ \mathrm{J}$ of work.

b. The system absorbs $1200\ \mathrm{J}$ of heat energy and receives $400\ \mathrm{J}$ of work.

c. The system expels $300\ \mathrm{J}$ of energy while its volume remains constant.

VI. (15 pts) An engine has a thermal efficiency of 40%. Calculate:

a. The work done by the engine if it absorbs $2000\ \mathrm{J}$ of heat energy from the hot reservoir.

b. The amount of heat energy expelled to the cold reservoir.

VII. (15 pts) An air-core solenoid has a length of $50\ \mathrm{cm}$ and a diameter of $3.0\ \mathrm{cm}$. It is wrapped with $3000\ \mathrm{turns}$. When a current of $5\ \mathrm{A}$ passes through it, calculate:

Given: The permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \mathrm{Tm/A}$.

a. The magnetic field inside the solenoid.

b. The length of the wire used to wrap the solenoid.

Exam Center: …… **Senior High School National Exam**

Room N°: …… **Seat N°:** …… **Examination Date: 24 August 2015**

Full Name: …… **Subject: Physics (Science)**

Signature: ……... **Duration: 90 minutes**

Score: 75 marks

Topics:

I. (8 pts) State the kinetic theory of an ideal gas.

II. (8 pts) What is the superposition of a wave?

III. (14 pts) Calculate the volume occupied by $2.8\ \text{g}$ of nitrogen gas in a container at a pressure of $1.0\times10^{5}\ \text{Pa}$ and a temperature of $300\ \text{K}$. Given: the molar mass of nitrogen, $M = 28\ \text{g/mol}$.

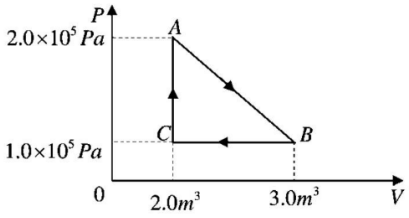
IV. (15 pts) A thermodynamic system undergoes the processes shown in the figure below. Calculate:

a. The work done during the process from A to B.

b. The work done during the process from B to C.

c. The work done during the process from C to A.

d. The net work done during the cyclic process ABCA.



V. (15 pts) A diesel engine in a vehicle has a thermal efficiency of 0.45 and absorbs $4.0\times10^{6}\ \text{J}$ of energy. Calculate:

a. The mechanical work produced by the piston.

b. The amount of energy expelled to the atmosphere.

c. The useful work output if the mechanical efficiency is 0.80.

VI. (15 pts) A copper wire with a diameter of $0.2\ \text{mm}$ and resistivity $\rho = 1.7\times10^{-8}\ \Omega\cdot\text{m}$ is wrapped into 6000 turns to form an air-core solenoid with a diameter of $3.0\ \text{cm}$ and a length of $60\ \text{cm}$. The solenoid carries a current of $1.0\ \text{A}$.

Given: The permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \text{Tm/A}$, calculate:

a. The magnetic field inside the solenoid.

b. The length of the wire used to wrap the solenoid.

c. The resistance of the wire.

Exam Center: …….

Room N°: …… **Seat N°:** ……

Full Name: …….

Signature: ……...

Score: 50 marks

Senior High School National Exam

Examination Date: 13 October 2014

Subject: Physics (Social)

Duration: 60 minutes

Topics:

I. (5 pts) Why is the box of a compass made from nickel or plastic instead of iron?

II. (7 pts) What is an internal combustion engine? What is an external combustion engine? Give an example for each.

III. (7 pts) What is sinusoidal motion? What is the result of the maximum displacement when two sinusoidal waves are out of phase?

IV. (7 pts) A coil has $N = 100$ turns and an average radius $R = 0.1$ m. It carries a current of $I = 10$ A. Calculate the magnetic induction B at the center O of the coil.

V. (12 pts) A Carnot engine performs 1500 J of work in one cycle, which lasts $t = 0.25$ s. The engine operates between a hot reservoir at temperature $T_h = 850$ K and a cold reservoir at temperature $T_c = 340$ K.

a. Calculate the efficiency of the engine.

b. Calculate the average power output of the engine.

c. Determine the amount of heat energy absorbed from the hot reservoir.

d. Determine the amount of heat energy rejected to the cold reservoir.

VI. (12 pts) An inductor of length $l = 0.50$ m and diameter $D = 4.0$ cm is closely wound with a wire of diameter $d = 1.0$ mm. The wire has resistivity $\rho = 2.5 \times 10^{-7}$ Ω·m.

a. Sketch (or describe) the direction of the electric current in the helical winding and the direction of the magnetic field $\vec{B}$ produced by the coil.

b. Calculate the resistance of the winding (neglect the thickness of the insulation).

Exam Center: …… **Senior High School National Exam**

Room N°: …… **Seat N°:** …… **Examination Date: 13 October 2014**

Full Name: …… **Subject: Physics (Science)**

Signature: ……... **Duration: 90 minutes**

Score: 75 marks **Second Attempt**

Topics:

I. (10 pts) What is the study of the first law of thermodynamics about? State the first law of thermodynamics.

II. (12 pts) Calculate the volume occupied by $9.6\ \text{g}$ of oxygen gas in a container at a pressure of $1.0\times10^{5}\ \text{Pa}$ and a temperature of $300\ \text{K}$. Given: the molar mass of oxygen, $M = 32\ \text{g/mol}$.

III. (12 pts) Calculate the change in internal energy ΔU of the system under the following conditions:

a. The system absorbs $500\ \text{calories}$ of heat energy and does $400\ \text{J}$ of work.

b. The system absorbs $300\ \text{calories}$ of heat energy and receives $420\ \text{J}$ of work from an external force.

c. The system expels $1200\ \text{calories}$ of heat energy while its volume remains constant.

Given: $1\ \text{calory} = 4.19\ \text{J}$.

IV. (13 pts) A gasoline engine receives $4.0 \times 10^{6}\ \text{J}$ of heat energy and has a thermal efficiency of 0.40. Calculate:

a. The mechanical work delivered by the piston.

b. The amount of heat energy expelled to the atmosphere.

c. The work done by the crankshaft if the mechanical efficiency is 0.85.

V. (13 pts) Two straight wires, each of length $l_1 = l_2 = 1\ \text{m}$, are placed parallel to each other in air and separated by a distance of $1\ \text{cm}$. Both wires carry currents of magnitude $I_1 = I_2 = 1\ \text{A}$ flowing in the same direction. The permeability of free space is $\mu_0 = 4\pi\times10^{-7}\ \text{Tm/A}$.

a. Calculate the magnitude of the force of interaction between the two wires.

b. Determine whether this force is attractive or repulsive.

VI. (15 pts) In an experiment to measure the magnitude of a magnetic field, electrons are accelerated from rest across an electric potential difference of $350\ \text{V}$. The magnetic field is perpendicular to the path of the electrons, causing them to move in a circular path with radius $R = 7.5\ \text{cm}$. Given: electron charge, $e = 1.6\times10^{-19}\ \text{C}$ and Electron mass, $m = 9.11\times10^{-31}\ \text{kg}$. Calculate:

a. The magnitude of the uniform magnetic field B.

b. The angular velocity of the electron, expressed in revolutions per second.

Exam Center: …… **Senior High School National Exam**

Room N°: …… **Seat N°:** …… **Examination Date: 13 October 2014**

Full Name: …… **Subject: Physics (Science)**

Signature: ……... **Duration: 90 minutes**

Score: 75 marks **First Attempt**

Topics:

I. (5 pts) What is the main purpose of using a transformer? Describe its structure.

II. (8 pts) When does the phenomenon of self-induction occur? Give an example to illustrate this phenomenon.

III. (12 pts) How do you use the right-hand rule to determine the direction of the magnetic field around:

a. A straight current-carrying wire?

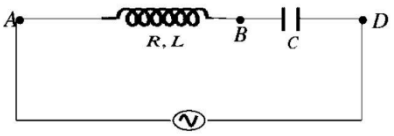
b. A circular loop of wire?

IV. (13 pts) A transformer has two coils: one with $10{,}000\ \text{turns}$ and another with $200\ \text{turns}$.

a. Which coil should be used as the primary coil if the goal is to increase the voltage?

b. Calculate the voltage across the terminals of the secondary coil if the voltage applied to the primary coil is $220\ \text{V}$.

V. (13 pts) An alternating current circuit has angular frequency ω (as shown). The r.m.s voltages and current are $V_{AB} = 160\ \text{V}$, $V_{BD} = 56\ \text{V}$, $V_{AD} = 120\ \text{V}$ and r.m.s current, $I = 0.2\ \text{A}$.



Calculate:

a) The impedances Z_{AB}, Z_{BD} and Z_{AD}.

b) The resistance of the coil.

VI. (15 pts) An inductor with inductance $L = 64\ \text{mH}$ is constructed using a conductor of diameter $d = 0.8\ \text{mm}$ covered by insulation of thickness $e = 0.1\ \text{mm}$. The wire is wound tightly in a spiral on a cylindrical core with diameter $D = 2\ \text{cm}$.

a. How many turns are needed to make this inductor?

b. To increase the inductance to $L_2 = 81\ \text{mH}$, the inductor is rewound using a thinner wire on the same cylinder. How many turns are required for this new inductor? Given: $\pi^2 = 10$

Exam Center: …….
Room N°: …… **Seat N°:** ……
Full Name: …….
Signature: ……...
Score: 50 marks

Senior High School National Exam
Examination Date: 05 August 2013
Subject: Physics (Social)
Duration: 60 minutes

Topics:

I. (5 pts) A container of volume $V = 0.5\times10^{-3}\ \mathrm{m}^3$ contains $n = 2.0$ moles of nitrogen gas under a pressure of $P = 6.0\ \mathrm{atm}$ $(1\ \mathrm{atm}=1.013\times105\ \mathrm{Pa})$. Calculate the average kinetic energy of each molecule of the gas under these conditions.

II. (7 pts) An AC source with an rms voltage of $V_{\mathrm{rms}} = 240\ \mathrm{V}$ and a frequency of $f = 50\ \mathrm{Hz}$ is connected in series with a capacitor of capacitance $C = 4700\ \mathrm{pF}$. Determine:

a. The capacitive reactance of the capacitor.
b. The rms current in the circuit.

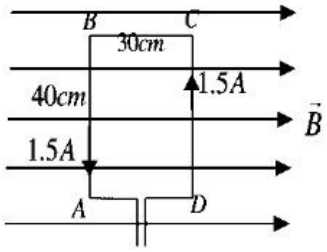
III. (14 pts) A Carnot engine receives $3352\ \mathrm{kJ}$ of heat from a hot reservoir at a temperature of $327\ ^\circ\mathrm{C}$. The engine performs work and rejects heat to a cold reservoir at a temperature of $127\ ^\circ\mathrm{C}$. Calculate:

a. The efficiency of the engine.
b. The work done by the engine.
c. The heat energy rejected to the cold reservoir.

IV. (12 pts). A person stands on a dock observing the motion of water waves that form sinusoidal waves. From this observation, the distance between two consecutive crests is $1.60\ \mathrm{m}$. Every $4.0\ \mathrm{seconds}$, a wave crest reaches the shore of the dock. Calculate the frequency and the speed of the wave.

V.(12 pts) A rectangular coil ABCD is placed in a uniform magnetic field of magnitude $B = 0.2\ \mathrm{T}$. The plane of the coil is parallel to the magnetic field lines. A current of $I = 1.5\ \mathrm{A}$ passes through the coil. Calculate:

a. The magnitude of the magnetic force acting on side AB.
b. The magnitude of the magnetic force acting on side CD.
c. The maximum torque exerted on the coil.



Exam Center: …….
Room N°: …… **Seat N°:** ……
Full Name: …….
Signature: ……...
Score: 75 marks

Senior High School National Exam
Examination Date: 05 August 2013
Subject: Physics (Science)
Duration: 90 minutes

Topics:

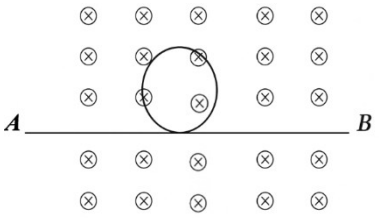
I. (5 pts) The molar mass of oxygen is $M = 32\ \mathrm{g/mol}$. Calculate the root-mean-square (rms) speed, v_{rms}, of oxygen molecules at a temperature of $T = 300\ \mathrm{K}$. Given the universal gas constant $R = 8.31\ \mathrm{J/mol.K}$.

II. (10 pts) The displacement of a sinusoidal wave traveling in the positive x-direction is given by: $y_1 = a\sin(kx - 2\pi ft)$ where a is the amplitude, f is the frequency, and $k = 2\pi/\lambda$ is the wave number. Another sinusoidal wave with the same amplitude and frequency propagates in the opposite direction and is described by: $y_2 = a\sin(kx + 2\pi ft)$.

a. Write the equation of the resultant wave formed by the superposition of y_1 and y_2.

b. Identify the positions of nodes and antinodes of the resultant wave.

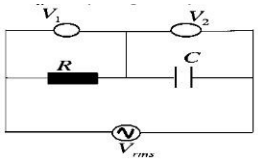
III. (15 pts) A uniform magnetic field $B = 25\ \mathrm{mT}$ is directed into the plane of the paper. A circular conducting loop with diameter $d = 2.0\ \mathrm{cm}$ lies in the plane of the paper.



a. The loop is rapidly tightened, causing its diameter to decrease to zero in a time interval of $\Delta t = 50\ \mathrm{ms}$. Calculate the induced electromotive force (emf) between terminals ***A*** and ***B***. Also, indicate the direction of the induced current.

b. Suppose the loop's diameter remains constant, but the magnetic field increases from $25\ \mathrm{mT}$ to $100\ \mathrm{mT}$ in a time interval of $\Delta t = 4\times10^{-3}\ \mathrm{s}$. Calculate the induced emf between terminals A and B, and indicate the direction of the induced current.

IV. (15 pts). A resistor of resistance $R = 500\ \Omega$ and a capacitor C are connected in series to an AC source with frequency $f = 50\ \mathrm{Hz}$. Two voltmeters with very high input impedance measure the rms voltages across the resistor V_1 and the capacitor V_2, respectively: $V_1 = 120\ \mathrm{V},\ V_2 = 160\mathrm{V}$, Calculate:



a. The rms current I passing through the circuit.

b. The power supplied by the source.

c. The capacitance C of the capacitor.

V.(15 pts) A monochromatic light source of wavelength $\lambda = 580\ \mathrm{nm}$ illuminates two slits S_1 and S_2 separated by a distance $a = 0.1\ \mathrm{mm}$. A screen is placed at a distance $d = 100\ \mathrm{cm}$ from the slits. Calculate:

a. The position of the third dark fringe and the third bright fringe with maximum amplitude relative to the central maximum.

b. The position of the fifth dark fringe and the fifth bright fringe relative to the central maximum.

c. The distance between adjacent fringes on the screen.

VI. (15pts) A Carnot engine operates between two thermal reservoirs at temperatures $T_{\mathrm{h}} = 850\,\mathrm{K}$ and $T_{\mathrm{c}} = 300\,\mathrm{K}$. The engine produces work $1200\,\mathrm{J}$ per cycle, and the duration of one cycle is $t = 0.25\,\mathrm{s}$. Calculate:

a. The efficiency of the engine.
b. The power output of the engine.
c. The heat energy absorbed from the hot reservoir per cycle.
d. The heat energy expelled to the cold reservoir per cycle.

Exam Center: ……. **Senior High School National Exam**

Room N°: …… **Seat N°:** …… **Examination Date: 06 August 2012**

Full Name: ……. **Subject: Physics (Social)**

Signature: ……... **Duration: 60 minutes**

Score: 50 marks

Topics:

I. (10 pts) A solenoid has a length $l = 50\ \text{cm}$, a total number of turns $N = 500$, and a cross-sectional area $A = \dfrac{100}{\pi}\ \text{cm}^2$. Given the permeability of free space $\mu_0 = 4\pi\times10^{-7}\ \text{T.m/A}$.

a. Calculate the inductance of the solenoid.

b. If the current through the solenoid changes at a rate of $\dfrac{di}{dt} = 2.0\ \text{A/s}$, calculate the induced electromotive force (emf) in the solenoid.

II. (10 pts) A vibration at the end of a rope A is described by the equation: $y_A = (4\ \text{cm})\ \sin(\pi t)$

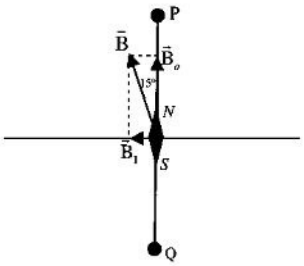
a. Calculate the frequency of the vibration.

b. Given that the vibration propagates along the rope with a speed of $v = 10\ \text{m/s}$, calculate the wavelength.

c. Write the wave equation at point O, which is located at a distance of ½ m from point A.

III. (15 pts) A copper wire PQ has a length of $6.0\ \text{m}$ and a diameter of $2.0\ \text{mm}$. It is stretched along a longitudinal magnetic field. At a distance of $10\ \text{cm}$ below this wire, a needle NS is placed, which can rotate in the horizontal plane about its center O.

A current I pass through the wire PQ in the direction from south to north, causing the needle to deflect by an angle of 15°.



Given: $\rho = 1.6\times10^{-8}\ \Omega\cdot\text{m}$, $B_0 = 2.0\times10^{-5}\ \text{T}$, $\mu_0 = 4\pi\times10\text{-}7\ \text{Tm/A}$, $\sin15° = 0.26$, $\cos 15° = 0.97$. Calculate:

a. The direction of deflection of the needle NS.

b. The current I passing through the wire PQ.

c. The potential difference between terminals P and Q.

IV. (15 pts). The temperature of a monoatomic ideal gas increases by $15\ \text{K}$ at constant volume. During this temperature change, the internal energy of the gas increases by $747.9\ \text{J}$. Given the gas constant $R = 8.31\ \text{J/mol.K}$.

a. Calculate the number of moles n of the gas.

b. Determine the work done by the gas during this process.

c. Calculate the amount of heat energy absorbed by the gas.

Exam Center: …….
Room N°: …… **Seat N°:** ……
Full Name: …….
Signature: ……...
Score: 75 marks

Senior High School National Exam
Examination Date: 06 August 2012
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (10 pts) A straight wire carries a current $I = 10\ \text{A}$. Calculate the magnetic field B at a point located $0.8\ \text{m}$ away from the wire. Given: $\mu_0 = 4\pi \times 10^{-7}\ \text{Tm/A}$.

II. (10 pts) A capacitor with capacitance $C = 2 \times 10^{-6}\ \text{F}$ is charged to a voltage $V = 2\ \text{V}$.

a. Calculate the energy stored in the capacitor.

b. The charged capacitor is then connected to an inductor with inductance L. If the maximum current passing through the inductor is $i_m = 2 \times 10^{-1}\ \text{A}$, calculate the inductance L.

III. (15 pts) An object undergoes two oscillations in the same direction and with the same frequency. The displacement equations are:
$y_1 = 5\sin(20t)$ and $y_2 = 5\sin(20t + \pi/2)$ where y_1 and y_2 are in centimeters and t is in seconds.

a. Write the equation for the resultant wave.

b. Calculate the amplitude of the resultant wave.

VI. (15 pts) A gas is enclosed in a cylinder with a cross-sectional area $A = 500\ \text{cm}^2$. The gas is confined by a movable piston on the top. Heat is supplied gradually to the gas while maintaining a constant pressure of $P = 10^5\ \text{Pa}$. As a result, the piston moves upward by a distance of $10\ \text{cm}$.

a. Identify the type of thermodynamic process occurring.

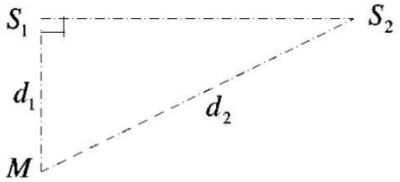
b. Calculate the work done by the gas during this process.

c. If the amount of heat supplied during this process is $600\ \text{J}$, calculate the change in the internal energy of the gas.

VII. (15 pts) Two wave sources S_1 and S_2 emit waves of the same frequency $f = 100\ \text{Hz}$. The distance between the sources is $S_1S_2 = 8\ \text{cm}$. The wave speed is $v = 1\ \text{m/s}$. These sources create an interference pattern.

a. Calculate the wavelength λ of the waves.

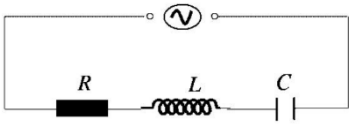
b. Calculate the path difference at point M between S_1 and S_2, where $S_1M = d_1 = 6\ \text{cm}$, and $S_1M \perp S_1S_2$ as shown in figure.



c. Determine the positions of point M where the wave amplitude is maximum and zero.

VIII. (15 pts) The diagram below shows a series RLC circuit consisting of a resistor R, an inductor L, and a capacitor C, connected to an AC source with an rms voltage $V_{rms} = 5\ \text{V}$ and frequency $f = 1\ \text{kHz}$. The rms voltages across the resistor, capacitor, and inductor are each $5\ \text{V}$, and the rms current in the circuit is $I_{rms} = 100\ \text{mA}$.

a. Calculate the resistance R, capacitance C, and inductance L of the components.



b. Calculate:

① The maximum voltage across the capacitor.

② The maximum charge stored on the capacitor.

③ The maximum voltage across the inductor and rate of change of current $\frac{di}{dt}$ in the inductor.

Exam Center: …….
Room N°: …… **Seat N°:** ……
Full Name: …….
Signature: ……...
Score: 50 marks

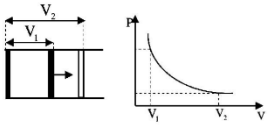
Senior High School National Exam
Examination Date: 27 July 2011
Subject: Physics (Social)
Duration: 60 minutes

Topics:

I. (10 pts) Calculate the root-mean-square (rms) speed of oxygen gas (O_2) at temperature $T = 0$ °C

Given: $R = 8.31$ J/mol. K, and molar mass of oxygen is $M = 32\times10^{-3}$ kg/mol.

II. (10 pts) An ideal gas of n moles is contained in a cylinder with a movable piston. The gas is supplied with heat Q and expands isothermally from volume V_1 to V_2 at constant temperature T, as shown in the figure. The work done by the gas during the expansion is 500 J.



a. What is the change in the internal energy of the system?
b. Calculate the heat supplied to the system.

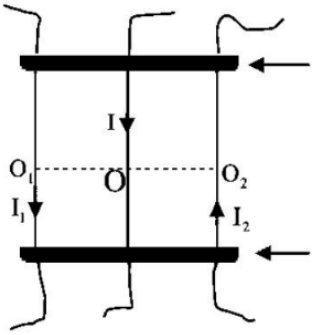
III. (15 pts) A Carnot engine performs $W = 2500$ J of work per cycle. It operates between a hot reservoir at $T_h = 600$ K and a cold reservoir at $T_c = 390$ K.

a. Calculate the efficiency of the engine.
b. Calculate the heat energy Q_h absorbed from the hot reservoir.

VI. (15 pts) A solenoid has a length $l = 25$ cm, radius $r = 2$ cm, and number of turns per unit length $n = 10^3$. A current passes through the solenoid, varying with time according to the relation: $i(t) = 10 - 2t$.

a. Calculate the inductance L of the solenoid. Take $\pi^2 = 10$.
b. Calculate the self-induced emf in the solenoid.

VII. (15 pts) Two identical straight long wires l_1 and l_2 of equal length are fixed and placed parallel to each other. The distance between the wires is $d = 0.8$ m. Both wires carry steady currents of equal magnitude flowing in opposite directions: $I_1 = I_2 = 2.0$ A. Point O lies midway between the two wires so that the perpendicular distances from O to l_1 and l_2 are both $d/2$.



a. Calculate the net magnetic induction at point *O* produced by the currents I_1 and I_2. (Give the magnitude and direction.)
b. A third straight wire of length $l = 1.0$ m is placed at point O, parallel to l_1 and l_2, carrying current $I = 10$ A in the direction shown. The wire at O is free to move. Calculate the net magnetic force exerted on this wire by l_1 and l_2. (Give magnitude and direction.)
c. Based on the forces, determine which direction the movable wire will be deflected.

Given: $\mu_0 = 4\pi\times10^{-7}$ Tm/A.

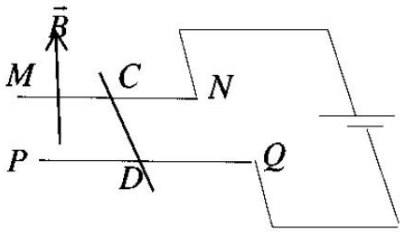
Exam Center: …….
Room N°: …… **Seat N°:** ……
Full Name: …….
Signature: ……...
Score: 75 marks

Senior High School National Exam
Examination Date: 27 July 2011
Subject: Physics (Science)
Duration: 90 minutes

Topics:

I. (10 pts) A solenoid has a total length of $l = 40.0$ cm and $N = 2000$ turns. Its total resistance is $R = 16.0\ \Omega$. When a constant voltage of $V = 20.0$ V is applied across the solenoid, determine the magnetic induction *B* at the midpoint along its axis. Given: $\mu_0 = 4\pi \times 10\text{-}7$ H/m.

II. (10 pts) Two long metal rails MN and PQ lie in a horizontal plane and are connected to a DC source. A conducting rod CD is placed across the rails so that it is perpendicular to them and can slide along them without friction. The plane $MNPQ$ lies in a uniform magnetic field $B = 0.20$ T directed perpendicular to the plane $MNPQ$. The rails are separated by a distance $d = 0.10$ m. A current $I = 10$ A flows through the rod and rails. Calculate:



a. The magnetic (Lorentz) force acting on the rod.
b. The mass of the rod, given that it accelerates with $a = 0.20$ m/s^2.

III. (15 pts) A transverse wave travels along a rope and is described by: $y\ (x,\ t) = 0.00327 \sin\ (72.1\ x - 2.72\ t)$ where y and x are in meters, and t is in seconds. The coefficients have the following units: 0.00327 m, 72.1 rad/m, 2.72 rad/s. Find:

a. The amplitude A of the wave.

b. The wavelength λ, period T, frequency *f*, and wave speed v.

VI. (15 pts) A solenoid has a length $l = 0.50$ m, number of turns $N = 500$, and a cross-sectional area $A = 20\times10^{-3}$ m^2. A time-varying current $i(t) = 10 \sin\ (100\pi t)$ (A) flows through the solenoid.

a. Determine the frequency of the current.
b. Calculate the inductance of the solenoid.
c. Write the expression for the self-induced electromotive force (emf) generated in the solenoid.

VII. (15 pts) A Carnot engine operates between a hot reservoir at temperature $T_h = 272$ °C and a cold reservoir at temperature T_c. The efficiency of the engine is estimated to be 0.20.

a. Calculate the temperature of the cold reservoir T_c in Kelvin.
b. Suggest ways to increase the efficiency of the engine and explain your reasoning.
c. Calculate the temperature of the cold reservoir if the engine's efficiency is increased to 40%

VIII. (15 pts) A straight wire of length $l = 1$ m moves with a speed $v = 0.025$ m/s for a duration of 2 seconds in a uniform magnetic field $\vec{B}$, where $B = 0.2$ T. The directions of B, $\vec{v}$, and the wire length $\vec{l}$ are mutually perpendicular.

a. Calculate the area swept out by the wire during the 2-second interval.

b. Calculate the magnetic flux passing through the area swept out by the wire.

c. Calculate the electromotive force (emf) induced in the wire.

d. Does the induced emf in the wire depend on the time duration of the displacement? Explain.